

Python Mini Project – Exploratory Data Analysis (EDA) & Visualization

DataSet :

https://drive.google.com/drive/folders/1pnk7VCENPr00_RRAisH9yXq1HYBVaxoR?usp=sharing

Note: Choose Dataset Wisely

Same Dataset for Excel, PowerBI, SQL and Python Project

Submission Link: <https://forms.gle/LinStt6ExLLP58Lu5>

Project Objective

To explore, understand, and visually analyse a real-world dataset using Python by applying exploratory data analysis techniques and appropriate data visualisation methods.

Project Instructions

1. You will be provided with a dataset in **CSV or Excel** format.
2. Perform the analysis using **Python (Jupyter Notebook preferred)**.
3. Use only **EDA and visualisation techniques**.
Do not build machine learning models.
4. The analysis must be **aligned with the actual structure of the dataset**.
5. If a specific column type is not available, adapt the analysis and clearly mention it.
6. Save the notebook as:
YourName_Python_EDA_Project.ipynb

Task 1: Import Libraries and Load Data

- Import required libraries:
 - pandas
 - numpy
 - matplotlib
 - seaborn

- Load the dataset into a pandas DataFrame.
- Display the first few rows.

What to show:

- Code cell loading the data
- Output showing sample rows

Task 2: Dataset Overview

- Display:
 - Shape of the dataset
 - Column names
 - Data types
- Use `.info()` and `.describe()` to summarise the data.

What to show:

- Output of dataset structure and statistical summary
- Short markdown note explaining the dataset size and types of variables

Task 3: Data Cleaning (EDA Level)

Perform basic cleaning required for analysis:

- Check for missing values and document them
- Handle duplicates if present
- Correct incorrect data types
- Standardise column names if necessary

What to show:

- Code and output for missing value and duplicate checks
- Brief explanation of cleaning decisions

Task 4: Univariate Analysis

Analyse individual variables based on their data type:

- **Numerical variables**
 - Histograms
 - Boxplots

- **Categorical variables**
 - Value counts
 - Bar plots

What to show:

- At least two plots
- Short interpretation for each plot

Task 5: Bivariate Analysis

Analyse relationships between variables:

- Numerical vs Numerical
 - Scatter plot
 - Correlation heatmap
- Categorical vs Numerical
 - Boxplot or bar chart

Select variable pairs based on availability in the dataset.

What to show:

- At least two bivariate plots
- Explanation of observed patterns or relationships

Task 6: Multivariate Analysis

- Use one of the following (as applicable):
 - Pairplot
 - Facet-based plots
 - Grouped bar charts
- Analyse interaction between more than two variables.

What to show:

- One multivariate visual
- Short explanation of insights

Task 7: Data Distribution & Outliers

- Identify skewness and spread of numerical variables

- Detect outliers using boxplots or IQR method

What to show:

- Visual evidence of distribution or outliers
- Interpretation of findings

Task 8: Visualization Summary

Create a **final visual summary**:

- 3–4 key plots that best describe the dataset
- Ensure plots have:
 - Titles
 - Axis labels
 - Proper scaling

Task 9: Key Insights

Write **3–5 concise insights** derived from the EDA and visualisations.

Examples:

- Patterns in distributions
- Strong or weak relationships between variables
- Presence of outliers or anomalies
- Differences across categories

Submission Requirements

- Jupyter Notebook (.ipynb)
- Well-commented code
- Clear markdown explanations
- Visualisations embedded in the notebook

Evaluation Criteria

- Correct and logical EDA flow
- Appropriate visual selection
- Alignment with dataset structure
- Quality of interpretation
- Code readability and organisation

